

■ Exercise Set 1: Understanding Symbolic Links (/bin, /sbin, /lib)

Exercise 1

Check whether /bin is a real directory or a symbolic link.
Find where it actually points to.

Solution:

```
ls -l /  
ls -l /bin
```

Output shows: /bin -> usr/bin

It is a symbolic link pointing to /usr/bin (usr-merge).

Exercise 2

List one command from /usr/bin.

Run that command using absolute path and normal name.

Solution:

```
ls /usr/bin | head -n 10  
/usr/bin/cat /etc/os-release  
cat /etc/os-release
```

■ Exercise Set 2: Important System Directories

Exercise 3 (/boot)

List files inside /boot.

Identify kernel-related files.

Solution:

```
ls /boot  
Kernel files: vmlinuz-*, initrd.img-*, config-*
```

Exercise 4 (/etc)

List files in /etc.

Open /etc/os-release.

Find OS name and version.

Solution:

```
ls /etc | head -n 20  
cat /etc/os-release
```

Exercise 5 (/var)

Navigate to /var/log.

List logs.

View last 10 lines.

Solution:

```
cd /var/log  
ls  
tail -n 10 auth.log
```

■ Exercise Set 3: User & Application Directories

Exercise 6 (/home)

```
cd ~  
mkdir linux_practice  
cd linux_practice  
touch test.txt
```

Exercise 7 (/root)

```
ls /root  
sudo ls /root
```

Exercise 8 (/opt)

```
ls /opt  
sudo mkdir /opt/myapp  
sudo touch /opt/myapp/version.txt
```

■ Exercise Set 4: Temporary & Virtual Filesystems

Exercise 9 (/tmp)

```
touch /tmp/testfile  
(Reboot clears /tmp)
```

Exercise 10 (/proc)

```
ls /proc
echo $$
cd /proc/$$
ls
```

```
Exercise 11 (/sys)
ls /sys/class
net, power_supply, block
```

```
Exercise 12 (/dev)
ls -l /dev/null
echo "Lost forever" > /dev/null
```

■ Exercise Set 5: Mount Points

```
Exercise 13 (/mnt)
ls /mnt
sudo mkdir /mnt/testmount
```

```
Exercise 14 (/media)
ls /media
(USB auto-mounts here)
```

```
Exercise 15 (/data)
cd /data
ls
touch newfile.txt
```